Code No: 115AN **R13**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year I Semester Examinations, January/February - 2023
PRINCIPLES OF PROGRAMMING LANGUAGES
(Computer Science and Engineering)

Time: 3 hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART - A

(25 Marks)

1.a)	Specify influences on Language design.	[2]
b)	Illustrate ambiguous grammar with an example.	[3]
c)	Write short notes on guarded commands.	[2]
d)	Compare record and union.	[3]
e)	What are called recursive subprograms?	[2]
f)	Write about generic subprograms in Ada and C++.	[3]
g)	List applications of Logic programming.	[2]
h)	Specify Inference process of prolog.	[3]
i)	Define data abstraction.	[2]
j)	List and summarize module libraries in python.	[3]

PART - B

(50 Marks)

2.a) Explain the process of compilation in each phase of a compiler.
b) Define BNF Grammar? How parse tree is generated for grammars? Explain with examples. [5+5]

OR

3.a) Describe the steps involved in the language evaluation criteria.
b) List and differentiate programming paradigms. [5+5]

4.a) Explain Categories of Arrays with suitable examples.
b) What do you mean by binding and binding time? Describe about various classes of binding times? [5+5]

OR

5.a) Discuss implementation of expressions and statements in C and C++ with examples.
b) Differentiate between static type checking and dynamic type checking and give their relative advantages. [5+5]

6.a) Illustrate stack based implementation of subprograms.
b) Elaborate design issues of subprograms and operations with example. [5+5]

OR

7.a) Explain about Co-Routines with an example.
b) Discuss about pass-by-result and pass-by-value-result parameter passing methods, with a detailed programming example for each. [5+5]

8.a) Illustrate concept of abstraction and encapsulation with suitable examples.
b) Elaborate how message passing helps in concurrency control? Explain with an example. [5+5]

OR

9.a) List and discuss different types of propositions present in logic programming.
b) Discuss about exception handling in Java with example. [5+5]

10.a) Discuss in detail about Scheme functional programming language.
b) Give a comparison between functional and imperative languages. [5+5]

OR

11.a) List and discuss in detail about data types in python with examples.
b) Explain the data objects in LISP. [5+5]

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